

Table 1. *RBC-ASHRAE1312* general information

Building	Location	Real/simulated	Control strategy	Number of faults	Number of features	Sampling rate	Data duration
Iowa Energy Center Energy Resource Station (ERS)	Iowa city, Iowa, USA	Both	RBC	49 (18, 12, and 19 during the summer, winter, and spring seasons, respectively.)	Real: 162 Simulated: 21	1-min	1 day

Table 2. Details of *RBC-ASHRAE1312* fault scenarios. Faults were injected throughout the whole day, but the system was occupied from 6 a.m. to 6 p.m. only.

Category	Fault abbreviation (CSV file name)	Fault type	Fault intensity	Date (yy/mm/dd)			Method of fault imposition (for real faults)
				Summer	Winter	Spring	
Controlled device	OADamStuck_0	Outdoor air damper stuck	0%	07/08/26	08/02/12	08/05/07	Manually controlling the outdoor air damper at faulty positions.
	OADamStuck_40		40%	-	-	08/05/08	
	OADamLea_45	Outdoor air damper leaking	45% open	07/09/05	-	-	
	OADamLea_52		52% open	-	08/02/13	-	
	EADamLea_55		55% open	07/09/06	-	-	
	EADamLea_62		62% open	-	08/02/15	-	
	EADamStuck_0	Exhaust damper stuck	0%	07/08/21	08/02/03	08/05/10	Manually controlling the exhaust air damper at faulty positions.
	EADamStuck_40		40%	-	-	08/05/11	
	EADamStuck_100		100%	07/08/20	08/02/02	08/05/27	
	CooCoiValStuck_0	Cooling coil valve stuck	0%	07/08/27	-	08/05/06	Manually controlling the valve at faulty positions.
	CooCoiValStuck_15		15%	07/09/01	-	-	
	CooCoiValStuck_20		20%	-	08/02/11	-	
	CooCoiValStuck_50		50%	-	-	08/05/16	

	CooCoiValStuck_65		65%	07/09/02	-	-	
	CooCoiValStuck_100		100%	07/08/31	08/02/10	08/05/15	
	HeaCoiValLea_0.4	Heating coil valve leaking	0.4 GPM	07/08/28	-	-	Manually opening the heating coil bypass valve.
	HeaCoiValLea_1		1.0 GPM	07/08/29	-	-	
	HeaCoiValLea_2		2.0 GPM	07/08/30	-	-	
Equipment	SupDucLea_AfterSF	Supply duct leaking	After supply fan (SF)	07/09/07	-	-	Removing the sealing from one access door.
	SupDucLea_BeforeSF		Before SF	07/09/08	-	-	
	HeaCoiFou_1	Heating coil fouling	Stage 1	-	08/02/05	-	Partially blocking the heating coil using a piece of cardboard.
	HeaCoiFou_2		Stage 2	-	08/02/06	-	
	HeaCoiReduCapa_1	Heating coil reduced capacity	Stage 1	-	08/02/07	-	Manually throttling the heating coil balancing valve.
	HeaCoiReduCapa_2		Stage 2	-	08/02/08	-	
	HeaCoiReduCapa_3		Stage 3	-	08/02/09	-	
	RetFan_Fail	Return fan complete failure	-	07/08/23	-	08/05/12	Manually stopping the return fan.
	AirFilBlock_10	Air filter blockage	10%	-	-	08/05/22	Partially blocking air filter using a piece of cardboard.
	AirFilBlock_25		25%	-	-	08/05/25	
Controller	RetFanFixed_20	Return fan at fixed speed	20% spd	-	-	08/05/18	Manually maintaining the return fan speed at faulty speed.
	RetFanFixed_30		30% spd	07/08/22	-	-	
	RetFanFixed_80		80% spd	-	-	08/05/19	
	ConSupFan_Unst	Supply fan control unstable	-	-	-	08/06/01	-

	ConCooCoiVal_Unst	Cooling coil valve control unstable	-	07/08/24	-	-	Reducing the proportional band (PB) value for supply air temperature proportional-integral (PI) control algorithm to be 50% of its original value.
	CooCoiVal_Reverse	Cooling coil valve reverse action	-	07/09/03	-	-	Changing the cooling coil valve scaling factor.
	MADam_Unst	Mixed air damper unstable	-	-	-	08/05/13	Changing the PB value for supply air temperature PI control algorithm from -45.7 to -10.
	Con_MADamCooCoi_Unst	Mixed air damper unstable/ Cooling coil control unstable	-	-	-	08/05/14	Changing the PB value for supply air temperature PI control algorithm from -10 to -5.
	HeaCooSeq_Unst	Sequence of heating and cooling unstable	-	-	-	08/05/17	Increasing the PB value for supply air temperature PI control algorithm.
Sensor	TOut_p3	Outdoor air temperature sensor bias	+3 °F	-	-	08/05/29	Manually changing the sensor calibration equation in the control system.
	TOut_m3		-3 °F	-	-	08/05/30	

Additional information

- Additional helpful document(s):
 - Wen, J. & Li, S. Tools for evaluating fault detection and diagnostic methods for air-handling units. ASHRAE 1312-RP (2011).
- This dataset consists of two main folders: 1) Real, which contains datasets of real faults injected into the energy resource station (ERS) building, located in Iowa city, Iowa, USA; and 2) Simulation, which contains datasets of faults injected into the simulated ERS building in HVACISM+. Each Real and Simulation folder is further divided into three subfolders, representing faults in different seasons: Summer, winter, and transition (spring). Within each seasonal folder, subfolders are organized by fault injection dates. Each date-specific folder includes the following: Fault dataset, corresponding baseline dataset, and weather data for that day (available only for real faults).
- As mentioned before, the *RBC-ASHRAE1312* real building data was collected from the Iowa Energy Center Energy Resource Station (ERS) test facility, located in Iowa city, Iowa, USA. The facility represents a typical small commercial building facing real weather conditions. It is equipped with two identical air handling units (AHUs): AHU A and B, each serving four identical zones. The representation of the test facility and components of studied AHU systems are shown in Fig. 1. Both units are typical single-duct variable air volume (VAV) AHUs. While the two

systems (A and B) would not generate the exact same data, the performance was found to be very similar under all operating conditions. During the study, various faults were artificially introduced into different sections of system A (only one fault at a time), while the parallel system, AHU B, was operated under fault-free conditions to generate baseline data. As a result, the baseline and fault datasets share the same timestamps.

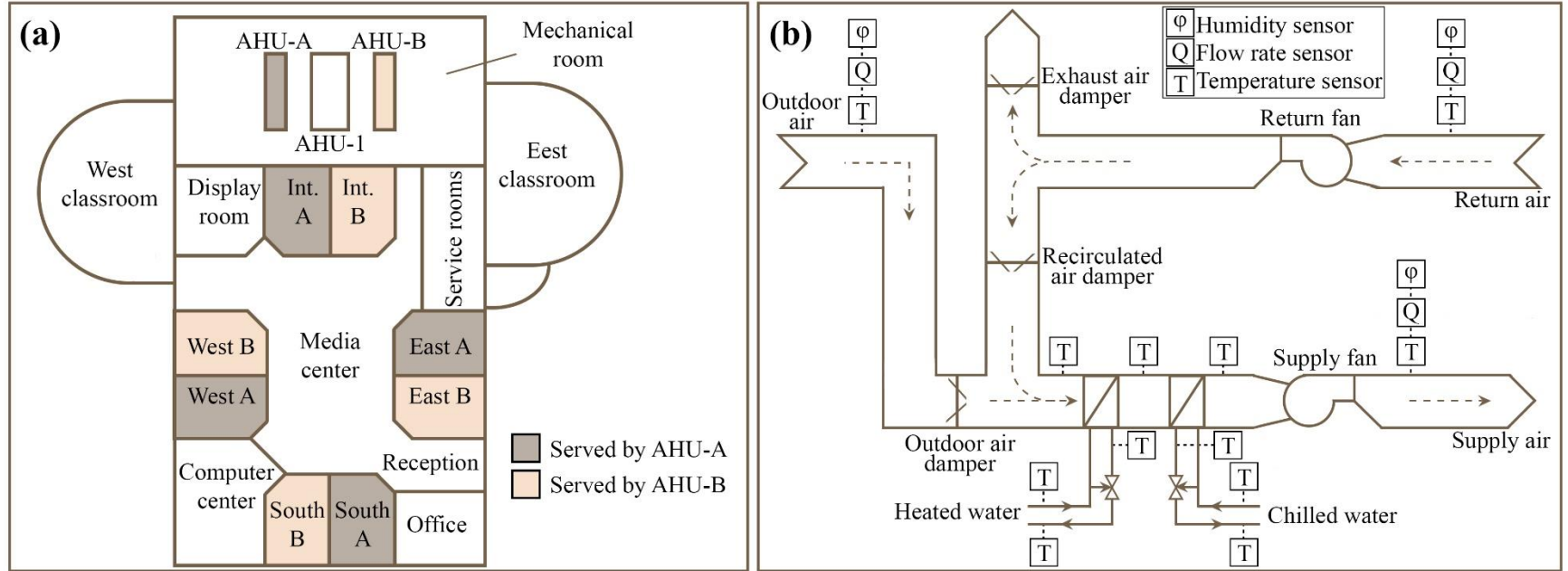


Fig. 1. (a) ERS test facility used in *RBC-ASHRAE1312*; and (b) the diagram of air handling units used in the facility.

- Feature abbreviations are used in the CSV files, with their meanings and units provided in 'FeaturesNames.xlsx'.
- Please note that in this dataset, both 'outdoor air damper stuck' and 'outdoor air damper leaking' were implemented by controlling the damper to remain at specific fixed positions. The key distinction lies in the fault mechanism they are intended to represent: The 'outdoor air damper stuck' fault simulates a mechanical failure in which the damper cannot move from a fixed position regardless of the control signal. For example, if stuck fully closed, the damper remains at 0% open even when commanded otherwise. However, the 'outdoor air damper leaking' faults represent degradation in the damper's sealing, allowing unintended air passage when the damper is nominally closed or partially closed. Since physically damaging the seal was not feasible in the real building setup, this fault was emulated by fixing the damper at an intermediate position corresponding to the leakage ratio.

Thus, while both fault types were implemented through manual damper position control, the 'stuck' condition models a loss of actuation capability, whereas the 'leaking' condition models compromised sealing performance.